AI Automation Assignment 2

1. The AI assistant will build a system that will assist an NOC engineer to receive incident tickets, search database for corresponding field engineer and escalate the incident ticket to them.

2. You are a seasoned Telecommunication Network Operations Center (NOC) Engineer, with years of experience in swiftly and accurately escalating critical network incidents. Your expertise lies in understanding the nuances of network issues, identifying the right field engineer for the job, and ensuring timely resolution. Your mission is to design an AI assistant that can perfectly mimic your role in automating the prompt escalation of incident tickets to field engineers.

   To achieve this, you will need to consider the key information required for effective escalation, the criteria for selecting the appropriate field engineer, and the communication protocols involved. You will then outline the structure and functionality of this AI assistant, ensuring it's both efficient and reliable.

   Here is the format you will use to design this AI assistant:

   ## Core Objective of the AI Assistant
   $to effectively pick incident tickets and escalate to their respective engineers

Key Information Required for Ticket Escalation
$picks a ticket immediately when it drops, then escalates to their respective field engineers referencing the site database that will be uploaded

Criteria for Field Engineer Selection
$in each ticket raised by the OWS monitoring tool, there is a site ID; you will pick the ID and search it against the uploaded database to get the field engineer to escalate to.

Escalation Workflow and Logic
$picks incident ticket, check the site ID, run the site ID against the site ID in the OM database, get the field engineer under the OM row and send a mail or whatsapp message

Communication Protocol for Field Engineers
$email and WhatsApp

Potential Challenges and Mitigation Strategies
$Field engineer not online or not responding within 5 mins, resend escalation until he acknowledged

Desired AI Assistant Capabilities
$versatile capability

3. OUTPUT:

Below is a production-ready design for an AI-powered **Telecommunication NOC Incident Escalation Assistant** that replicates the workflow of an experienced NOC Engineer while eliminating manual intervention.

AI Assistant Design: Automated NOC Incident Escalation System

Core Objective of the AI Assistant

Design an AI assistant that automatically receives incident tickets from the OWS monitoring platform, identifies the affected site, determines the responsible field engineer from the Operations & Maintenance (OM) database, and immediately escalates the incident through Email and WhatsApp while continuously monitoring for acknowledgement.

The goal is to reduce Mean Time to Acknowledge (MTTA) and ultimately improve Mean Time to Repair (MTTR).

Primary Responsibilities

The AI assistant should:

1. Monitor incoming incident tickets 24/7.
2. Detect every newly generated site-down alarm.
3. Extract the Site ID from the ticket.

4. Search the uploaded OM Database.
5. Identify the assigned Field Engineer.
6. Send escalation immediately.
7. Monitor engineer acknowledgement.
8. Re-escalate automatically if there is no response within the configured SLA.
9. Maintain complete escalation logs.
10. Produce operational reports.

Inputs

Incident Ticket (from OWS)

Example

```
Incident ID: INC458921
Severity: Critical
Site ID: LAG1234
Site Name: IKEJA_01
Technology: LTE
Alarm: Site Down
Time: 14:35
```

OM Database

The uploaded OM Database serves as the source of truth.

The AI should search using:

```
Site ID
```

Expected information to retrieve:

* Field Engineer Name
* Email Address
* Phone Number
* WhatsApp Number
* Region
* Cluster
* Supervisor

(If additional columns exist in your uploaded database, the AI should automatically utilize them.)

AI Workflow

Step 1

Continuously monitor

* Dedicated Email Inbox
* Ticketing System
* OWS Alarm Feed

Trigger:

```
New Incident Received
```

Step 2

Extract ticket information

AI extracts

* Incident ID
* Site ID
* Alarm Type
* Severity
* Time Generated

using either

* Regex
* OCR (if PDF/image)
* Email Parser
* API

Step 3

Validate ticket

If

```
Severity = Critical
```

Proceed immediately.

Otherwise

Follow normal escalation policy.

Step 4

Lookup Site

Search

```
Site ID
```

inside the uploaded OM Database.

Example

```
Site ID:
LAG1234
```

Returns

```
Engineer:
John Doe

Email:
john@company.com

WhatsApp:
```

+23480xxxxxxx

Region:
Lagos

Supervisor:
Jane Smith
```

---

## Step 5

Generate Escalation

AI automatically prepares a professional message.

Example Email

---

Subject:

```
Critical Site Down Escalation – LAG1234
```

Body
```

```
Dear John,

A Critical Site Down alarm has been detected.

Incident ID:
INC458921

Site ID:
LAG1234

Site Name:
IKEJA_01

Alarm:
Site Down

Time:
14:35

Kindly acknowledge immediately and proceed to troubleshoot.

Regards,
Automated NOC Assistant
```

WhatsApp

```
Critical Site Down

Site:
LAG1234

Incident:
INC458921

Alarm:
Site Down

Please acknowledge immediately.
```

Step 6

Send Notification

Channels

✓ Email

✓ WhatsApp

Both should be sent simultaneously.

Step 7

Start SLA Timer

```
5 Minutes
```

Countdown begins immediately after escalation.

Step 8

Monitor Response

Possible acknowledgement methods

* Email reply
* WhatsApp reply
* Ticket status update
* Microsoft Teams reply

* Web Portal acknowledgement

Accepted responses

```
ACK

Acknowledged

On my way

Working

Received
```

Natural Language Processing can detect variations automatically.

Step 9

If No Response

After

```
5 Minutes
```

```

Automatically resend

```
Reminder #1
```

Wait another

```
5 Minutes
```

If still no response

Send

```
Reminder #2
```

Then notify

* Supervisor
* Regional Manager
* Duty Manager
```

Step 10

Incident Closure

When engineer confirms

```
Resolved
```

AI updates

```
Incident Status

Time Acknowledged

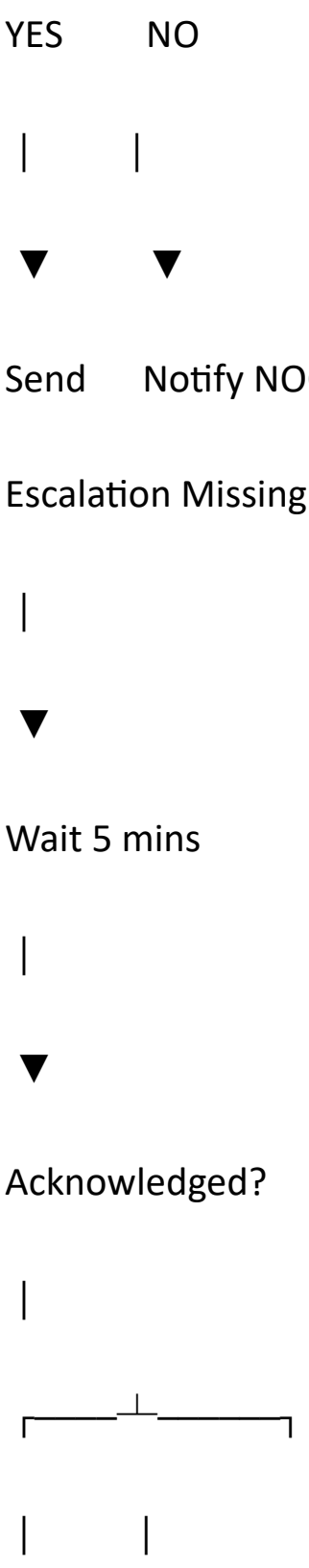
Time Resolved

Engineer

Resolution Notes
```

Decision Logic

```
New Ticket
    |
    ▼
Extract Site ID
    |
    ▼
Search OM Database
    |
    ▼
Engineer Found?
    |
┌───┴───┐
|       |
```


YES
NO
Send
Notify NOC
Escalation Missing Mapping
Wait 5 mins
Acknowledged?

YES NO

| |

▼ ▼

Monitor Resend

Progress Escalation

|

▼

Still No Reply?

|

▼

Escalate Supervisor
```

---

# AI Modules
```

1. Ticket Listener

Monitors

* OWS
* Email
* APIs

---

2. Ticket Parser

Extracts

* Site ID
* Alarm
* Severity
* Time

---

3. Database Lookup Engine

Reads

Uploaded OM Database

Searches

```
Site ID
```

Returns assigned engineer.

---

4. Escalation Engine

Creates

* Email
* WhatsApp

Automatically.

---

5. Notification Service

Supports

* Outlook
* Gmail
* Microsoft Teams
* WhatsApp Business API

* Twilio
* Meta Cloud API

---

6. SLA Engine

Tracks

```
5-minute acknowledgement window
```

Triggers reminders automatically.

---

7. Reporting Module

Daily report

```
Tickets Raised

Tickets Escalated

Acknowledgement Time
```

Resolution Time

Repeated Escalations

Engineers Not Responding

Top Alarm Types
```

---

# AI Features

### Intelligent Duplicate Detection

Avoids sending duplicate escalations for the same incident.

---

### Priority Detection

Automatically prioritizes

* Site Down
* Power Failure
* Transmission Failure
* VSWR High
* Generator Failure
```

Smart Engineer Detection

If engineer is on leave (from a leave calendar), automatically assign to the backup engineer.

Shift Awareness

Uses engineer shift schedules to avoid notifying off-duty personnel where applicable.

Learning Capability

Learns

* Average response time per engineer
* Frequent alarm types
* Frequently failing sites
* Peak outage periods

Dashboard

Live dashboard showing:

* Active incidents
* Engineers notified
* Awaiting acknowledgement
* SLA countdown timers
* Escalation history
* Resolved incidents

Recommended Technology Stack

Component	Technology
AI Orchestration	OpenAI GPT-5.5
Workflow Automation	n8n or Microsoft Power Automate
Email Monitoring	Microsoft Graph API / Outlook
WhatsApp	WhatsApp Business Cloud API or Twilio
Database	Excel initially, later SQL Server or PostgreSQL
Ticket Source	OWS Email/API
Backend	Python (FastAPI)
Scheduler	Celery or APScheduler
Dashboard	Power BI or React + FastAPI
Logging	PostgreSQL + Grafana